Problem Solving Testing Using Java (PSTJ)

Introduction :

This repository represents a comprehensive collection of structured programming tasks designed as part of Practical Software Training and Java coursework. The content is carefully organized to guide learners from basic programming concepts to advanced problem-solving and system-level thinking.

The curriculum is divided into 5 units. Each unit spans 3 weeks, and every week consists of 10 tasks. This results in a total of 150 tasks, ensuring consistent practice and gradual skill development.

Objectives :

The primary objectives of this repository are:

* To build a strong foundation in programming concepts
* To enhance logical thinking and analytical skills
* To provide hands-on experience with Java programming
* To introduce real-world problem-solving approaches
* To prepare learners for technical interviews and academic evaluations

Repository Structure :

id=”structure”

```
PSTJ/
│
├── Unit1/
│   ├── Week1/
│   │   ├── Task1.java
│   │   ├── Task2.java
│   │   └── ... Task10.java
│   ├── Week2/
│   ├── Week3/
│
├── Unit2/
│   ├── Week1/
│   ├── Week2/
│   ├── Week3/
│
├── Unit3/
│   ├── Week1/
│   ├── Week2/
│   ├── Week3/
│
├── Unit4/
│   ├── Week1/
│   ├── Week2/
│   ├── Week3/
│
```

```
├── Unit5/
│   ├── Week1/
│   ├── Week2/
│   ├── Week3/
│
└── README.md
```

Learning Approach :

This repository follows a progressive learning methodology:

* Concept Introduction: Each week starts with basic problems to introduce concepts
* Practice and Reinforcement: Mid-level tasks strengthen understanding
* Advanced Application: Final tasks challenge the learner with complex scenarios

The tasks are designed to promote independent thinking and practical implementation rather than rote learning.

Unit-wise Breakdown

Unit 1: Fundamentals of Programming

* Introduction to Java syntax

* Variables, data types, and operators
* Conditional statements and loops
* Basic input/output handling

Unit 2: Control Structures and Methods

* Functions and modular programming
* Nested loops and decision-making
* Code reusability and readability
* Basic debugging techniques

Unit 3: Data Structures and Algorithms

* Arrays and strings
* Searching and sorting techniques
* Time and space complexity basics
* Problem-solving strategies

Unit 4: Object-Oriented Programming

* Classes and objects
* Inheritance, polymorphism, encapsulation
* Exception handling
* File handling basics

Unit 5: Advanced Concepts and Applications

* Collections framework
* Multithreading basics
* Performance optimization
* Real-world application scenarios

Weekly Workflow

Each week is structured as follows:

* Task 1–3: Basic concept implementation
* Task 4–6: Intermediate problem-solving
* Task 7–10: Advanced and real-world scenarios

Suggested workflow:

1. Read and understand the problem
2. Design the logic or algorithm
3. Implement the solution
4. Test with multiple inputs
5. Optimize if necessary

Coding Guidelines

To maintain consistency across the repository:

* Use meaningful variable and method names
* Follow proper indentation and formatting
* Include comments for important logic
* Avoid unnecessary complexity
* Ensure code readability

Tools and Technologies

* Programming Language: Java
* Recommended IDEs:

  * IntelliJ IDEA
  * Eclipse
  * Visual Studio Code

Prerequisites

Before starting, you should have:

* Basic understanding of programming concepts
* Familiarity with Java syntax (optional for beginners)
* Installed Java Development Kit (JDK)

Best Practices

* Practice regularly instead of completing tasks in bulk
* Try solving problems without referring to solutions first
* Review and refactor your code
* Maintain a record of mistakes and improvements
* Focus on understanding rather than memorizing

Common Challenges

While working through the tasks, we may encounter:

* Logical errors in problem-solving
* Difficulty in debugging code
* Performance issues in larger problems
* Understanding advanced concepts

These challenges are part of the learning process and help in skill development.

Conclusion

This repository serves as a complete practice resource for mastering Java programming through structured and progressive learning. By consistently working through the tasks, learners can develop strong programming skills and confidence in solving real-world problems.